A few sample problems focusing on derivatives of quadratic forms and related matrix calculus concepts

---

Problem 1: Derivative of a Scalar Function of a Vector

Let:

$$f(\mathbf{x}) = \mathbf{x}^T\mathbf{x}$$

Where:

- $\mathbf{x} \in \mathbb{R}^n$

Question:

Find $\frac{df}{d\mathbf{x}}$

Problem 2: Derivative of Quadratic Form with Non-Symmetric Matrix

Let:

$$f(\mathbf{w}) = \mathbf{w}^T\mathbf{A}\mathbf{w}$$

Where:

- $\mathbf{w} \in \mathbb{R}^n$
- $\mathbf{A} \in \mathbb{R}^{n\times n}$ is *not necessarily* symmetric

Question:

Find $\frac{df}{d\mathbf{w}}$

Problem 3 : Derivative of a Linear Form

Let:

$$f(\mathbf{w}) = \mathbf{a}^T \mathbf{w}$$

Where:

- $\mathbf{w} \in \mathbb{R}^n$
- $\mathbf{a} \in \mathbb{R}^n$ is a constant vector

Question:

Find $\frac{df}{d\mathbf{w}}$

Problem 4: Derivative with Respect to a Matrix

Let:

$$f(\mathbf{X}) = \text{tr}(\mathbf{A}\mathbf{X})$$

Where:

- $\mathbf{X} \in \mathbb{R}^{n \times m}$
- $\mathbf{A} \in \mathbb{R}^{m \times n}$

Question:

Find $\frac{df}{d\mathbf{X}}$

A few sample problems focusing on gradients of common loss functions and related matrix calculus:-

Problem 1:

Let:

- $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$ where $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $\mathbf{w} \in \mathbb{R}^d$
- $L = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$ (Sum of Squared Errors, without the $\frac{1}{n}$ factor)

Question:
Derive the gradient $\frac{\partial L}{\partial \mathbf{w}}$

Problem 2: Gradient of Mean Absolute Error (MAE) Loss

Let:

- $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w} + \mathbf{b}$ where $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{w} \in \mathbb{R}^d$, and $\mathbf{b} \in \mathbb{R}^n$ (bias vector)
- $L = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_i - y_i|$

Question:
Derive the gradient $\frac{\partial L}{\partial \mathbf{w}}$ (You might need to consider subgradients for points where $|\cdot|$ is not differentiable).

Problem 3: Gradient with respect to Bias Term

Let:

- $\hat{Y} = XW + b$ where $X \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}$ (scalar bias, broadcast to all elements)
- $L = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$

Question:
Derive the gradient $\frac{\partial L}{\partial b}$

Problem 4 (Moderate): Gradient with L2 Regularization

Let:

- $\hat{Y} = XW$ where $X \in \mathbb{R}^{n \times d}$
- $L = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 + \lambda ||W||_F^2$ where $||W||_F^2 = \sum_{j=1}^{d} \sum_{k=1}^{m} W_{jk}^2$ (Frobenius norm squared) and λ is a regularization parameter. Assume $W \in \mathbb{R}^{d \times 1}$ for simplicity, making $||W||_F^2 = \mathbf{w}^T \mathbf{w}$.

Question:
Derive the gradient $\frac{\partial L}{\partial W}$